Appendix A Lineage-Linked GEDCOM Tag Definition

Introduction

Appendix A is a glossary of the tags used in the Lineage-Linked Form. These tags are used in a hierarchical structure to describe individuals in terms of their families, names, dates, places, events, roles, sources, and relationships. Control information and other kinds of data intended for computer processing is also included.

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Extending the Form

Records defined in the Lineage-Linked Form cannot not be redefined.
It is legal to extend the Lineage-Linked Form, but only by using user-defined tags which must begin with an underscore. The meaning of GEDCOM tags is defined by their context, e.g. HEAD.SOUR and FAM.SOUR are two different kind of sources. Software developers *are* allowed to introduce _TAG.SOUR; as that does not redefine any existing SOUR tag in any way. However, developers are disallowed from directly extending the Lineage-Linked Form using existing tags, so they are not allowed to introduce say OBJE.SOUR, but would have to use OBJE._SOUR instead.

A GEDCOM reader *should not* assume that predefined tags occur only in predefined context, but only assume that the records defined in the Lineage-Linked Form are as defined.

Illegal Tags

User-defined tags that do not start with an underscore are illegal.
GEDCOM readers that come across an illegal tag should issue a fatal error and abort. It may seem harsh to reject a GEDCOM file for one illegal tag, but there simply is no reason for a GEDCOM file to contain illegal tags. It's a trivial effort for the developer of the offending application to use underscores.

The obvious exception to this rule is a new version of the offending application reading an GEDCOM file produced by an earlier version of that application; applications should be able to read files produced by earlier versions of the same application, even if these files should otherwise be rejected.

FTW TEXT

The list of definitions below gives both the tag name and, within curly brackets, a longer text. Typically, the longer text is a valid English word, or two words connected by an underscore, and the tag an abbreviation of that word or phrase.

GEDCOM uses the tags, most of which are four code units long. Exactly one genealogy application decided to use the text within the curly brackets instead. That application is Family Tree Maker for Windows (FTW), and those files are not GEDCOM files. These file are known as FTW TEXT files.

Back when Family Tree Maker for Windows (Classic) was quite popular, these files used to be quite prevalent. There are few FTW TEXT files around any more. They can be converted to GEDCOM files using a freely downloadable copy of Family Tree Maker 2005 Starter Edition. A download link to the Family Tree Maker 2005 Starter Edition is provided in the Family Tree Maker support article *Opening Old and Unsupported Files in FTM 2008-2017 for Windows*. So, there is no need to support

FTW TEXT in modern genealogy applications.

FTW TEXT Detection

Detecting FTW TEXT files is easy.
While the first line of a GEDCOM file is `0 HEAD`, the first line of an FTW TEXT is `0 HEADER`.

GEDCOM Reader Best Practice

- do *not* add support for FTW TEXT
- **do detect** FTW TEXT, and when detected
  - inform the user that it isn't a GEDCOM file, but an FTW TEXT file
  - advise the user to convert the FTW TEXT to GEDCOM using Family Tree Maker 2005 Starter Edition
- if you *do* support FTW TEXT
  - do not do so silently, do not let the user keep thinking it's a GEDCOM file when it isn't
  - clearly state that you detected FTW TEXT, and are processing their FTW TEXT file
  - make sure you support both GEDCOM and FTW TEXT, but not FGTEWDTCEOXMT (a mixture of both); keep detecting illegal tags

A download link for Family Tree Maker 2005 Starter Edition is available in the MacKiev support item *Family Tree Maker support: Opening Old and Unsupported Files in FTM 2008-2017 for Windows*.

References

- FTW TEXT
- Family Tree Maker support: Opening Old and Unsupported Files in FTM 2008-2017 for Windows

Lineage-Linked GEDCOM Tag Definitions

This section provides the definitions of the Lineage-Linked GEDCOM tags and shows their formal name inside of the {braces}. The formal names are *not* used in place of the tag. Full understanding must come from the context in which the tag is used.

ABBR {ABBREVIATION}:=
A short name of a title, description, or name.

ADDR {ADDRESS}:=
The contemporary place, usually required for postal purposes, of an individual, a submitter of information, a repository, a business, a school, or a company.

ADR1 {ADDRESS1}:=
The first line of an address.

ADR2 {ADDRESS2}:=
The second line of an address.

ADR3 {ADDRESS3}:=
The third line of an address.

ADOP {ADOPTION}:=

Adoption is a legal event that changes a child's legal parents from one set of parents to another set of parents.
While some of the parents involved are likely to be biological or official parents, neither assumption should be made.

Adoption is an event that changes who the legal parents are. A child that has been adopted can be adopted again. The official parents should not be assumed to be the biological parents. A child can *and often is* adopted by a biological or official parent. In many jurisdictions, a child is technically always adopted by a couple, even if one of them is already is a legal parent.

AGE {AGE}:=
The age of the individual at the time an event occurred, or the age listed in the document.

AGNC {AGENCY}:=
The institution or individual having authority and/or responsibility to manage or govern.

ANUL {ANNULMENT}:=
Declaring a marriage void from the beginning (retroactively invalid).

ASSO {ASSOCIATES}:=
An indicator to link friends, neighbours, or associates, who aren't close relatives of an individual.

AUTH {AUTHOR}:=
The name of the individual who created or compiled information.

BAPM {BAPTISM}:=
The event of baptism, performed in infancy or later. (See also CHR, page 124.)

BARM {BAR_MITZVAH}:=
The religious ceremony held when a Jewish boy reaches age 13.

BASM {BAS_MITZVAH}:=
The religious ceremony held when a Jewish girl reaches age 13, also known as "Bat Mitzvah."

BIRT {BIRTH}:=
The emergence of offspring from their mother as a separate being.
Birth does not imply life. Birth includes stillbirth.

BURI {BURIAL}:=
The action of burying a body.
BURI includes all forms of burial, including burial at sea, and as there is no separate event for interment (entombment), BURI is used for that too.

CALN {CALL_NUMBER}:=
The number used by a repository to identify the specific items in its collections.

CAST {CASTE}:=
The name of an individual's rank or status in society which is sometimes based on racial or religious differences, or differences in wealth, inherited rank, profession, occupation, etc.

CAUS {CAUSE}:=
A description of the cause of the associated event or fact, such as the cause of death.

CENS {CENSUS}:=
The event of the periodic count of the population for a designated locality, such as a national or state Census.

CHAN {CHANGE}:=
Indicates a change, correction, or modification. Typically used in connection with a DATE to specify when a change in information occurred.

CHIL {CHILD}:=
The biological, official or legal (adopted) child of a parent or parents.

CHR {CHRISTENING}:=
The religious event of baptising *and* naming a child.

CHRA {ADULT_CHRISTENING}:=
The religious event of baptizing *and* naming an adult person.

CITY {CITY}:=
A lower level jurisdictional unit. Normally an incorporated municipal unit.

CONF {CONFIRMATION}:=
The religious rite that confirms membership of a church (*confirms* because previously established by baptism).

COPR {COPYRIGHT}:=
A statement that accompanies data to protect it from unlawful duplication and distribution.

CORP {CORPORATE}:=
A name of an institution, agency, corporation, or company.

CREM {CREMATION}:=
Disposal of a body by fire, by burning it to ashes.

CTRY {COUNTRY}:=
The name of the country.

DATA {DATA}:=
Data.

DATE {DATE}:=
The time of an event in a calendar format.

DEAT {DEATH}:=
The end of a life.

DEST {DESTINATION}:=
A system receiving data.

DIV {DIVORCE}:=
The legal dissolution of a marriage.

DIVF {DIVORCE_FILED}:=
An event of filing for a divorce by a spouse.

DSCR {PHYSICAL_DESCRIPTION}:=
The physical characteristics of a person, place, or thing.

EDUC {EDUCATION}:=
Indicator of a level of education attained.

EMAIL {EMAIL}:=
An electronic mail address.
GEDCOM 5.5.1 lists EMAIL as EMAI (no L) in this Appendix. A forgiving GEDCOM 5.5.1 reader may treat EMAI as a synonym for EMAIL.
A GEDCOM 5.5.5 writer *must* use EMAIL (the actual tag). A GEDCOM 5.5.5 reader *must* reject EMAI (no L) as an illegal tag and abort processing.

EMIG {EMIGRATION}:=
An event of leaving one's homeland with the intent of residing elsewhere.

ENGA {ENGAGEMENT}:=
An event of recording or announcing an agreement between two people to become married.

EVEN {EVENT}:=
Pertaining to a noteworthy happening related to an individual, a group, or an organisation.
An EVEN (event) structure is usually qualified or classified by a subordinate use of the TYPE record.

FACT {FACT}:=
Pertaining to a noteworthy attribute or fact concerning an individual, a group, or an organisation.
A FACT structure is usually qualified or classified by a subordinate use of the TYPE record.

FAM {FAMILY_GROUP}:=
The FAM (family group) structure records a single family group; a couple and their children. The group consist of two partners, either or both of which may be unknown, with or without children. The partners may or may not be spouses, and may or may not have children, but are biological, official or legal parents to each of the children in the group.
Recording a single family often requires more than one family group record.

FAMC {FAMILY_CHILD}:=
Identifies the family group in which an individual appears as a child.

FAMS {FAMILY_SPOUSE}:=

Identifies the family group in which an individual appears as a partner.

The name and abbreviation of this record are misleading: the individual need not be a spouse in that family group. The family group record is used for all relationships, not just marriages. *Do not assume* that the person is a spouse in that family group.

FAX {FACSIMILE}:=
Electronic facsimile transmission.

FCOM {FIRST_COMMUNION}:=
Literally the first communion an individual partakes in. Communion is a rite within christian churches, and the first communion is considered a rite of passage.

FILE {FILE}:=
The name of an external file, or, in the case of HEAD.FILE, the original filename of this GEDCOM file.

FONE {PHONETIC}:=
A phonetic rendering of a superior text string.

GIVN {GIVEN_NAME}:=
A given or earned name used for official identification of a person.

GRAD {GRADUATION}:=
An event of awarding educational diplomas or degrees to individuals.

HUSB {HUSBAND}:=
A partner in a FAM (family group) record, often male, often partner to a woman, and a biological, official or legal parent to each of the children of the couple.
The name of this record strongly suggests that the line value must identify a husband, but that is not the case; the relationship need not be a marriage, and the individual need not be male, it may identify a woman in a lesbian relationship.

References

Same-Sex Marriage in GEDCOM

IDNO {IDENT_NUMBER}:=

An identifier, often called a number, assigned to identify a person within some significant external system.
The value typically isn't number, but a value containing spaces and dashes in addition to letters and digits.
The prime example is a passport “number”. American genealogists often record Social Security Numbers.

The difference between the INDO record and the REFN record is that the IDNO record is for third-party numbers, and the REFN record is for user-defined numbers.

IMMI {IMMIGRATION}:=

An event of entering into a new locality with the intent of residing there.

INDI {INDIVIDUAL}:=

A person.

LANG {LANGUAGE}:=

The name of the language used in a communication or transmission of information.

LATI {LATITUDE}:=

Latitude of a position on the globe.

LONG {LONGITUDE}:=

Longitude of a position on the globe.

MAP {MAP}:=

Pertains to a representation of measurements usually presented in a graphical form.

MARB {MARRIAGE_BANN}:=

An event of an official public notice given that two people intend to marry.

MARC {MARR_CONTRACT}:=

An event of recording a formal agreement of marriage, including the prenuptial agreement in which marriage partners reach agreement about the property rights of one or both, securing property to their children.

MARL {MARR_LICENSE}:=

An event of obtaining a legal license to marry.

MARR {MARRIAGE}:=

Marriage is an official and legal event, defined by the applicable law and customs of the land and the time, that creates a couple, possibly with children. This includes so-called common law marriages.

The name of this record is ill-chosen. The MARR record isn't a marriage record, but a relationship record.
The MARR record can and must be used for all relationship types, with marriage merely being the default relationship type for the couple.

References

- Marriage in GEDCOM
- Same-Sex Marriage in GEDCOM

MARS {MARR_SETTLEMENT}:=
An event of creating an agreement between two people contemplating marriage, at which time they agree to release or modify property rights that would otherwise arise from the marriage.

MEDI {MEDIA}:=
Information about the media or having to do with the medium in which information is stored.

NAME {NAME}:=
Depending on context, a product name, repository name or an individual's full name.
The NAME *must not* contain nobility titles. The NAME *may* contain earned titles and salutations (See NPFX, page 127).
More than one NAME record should be used for individuals known by multiple names.

NATI {NATIONALITY}:=
The nationality of an individual.

NATU {NATURALISATION}:=
The event of obtaining citizenship.

NCHI {NUMBER_OF_CHILDREN}:=
The number of children that this individual (INDI.NCHI) or couple (FAM.NCHI) has.

NICK {NICKNAME}:=
A descriptive or familiar that is used instead of, or in addition to, one's proper name.

NMR {NUMBER_OF_RELATIONSHIPS}:=
The number of relationships (FAM records as a partner) this person would occur in if all relationships were included.

NOTE {NOTE}:=
Additional information provided by the submitter for understanding the enclosing data.

NPFX {NAME_PREFIX}:=
Text which appears on a name line before the given and surname parts of a name.
e.g. **Lt. Cmndr.** Joseph /Allen/ Jr.
In this example Lt. Cmndr. is considered as the name prefix portion.

NSFX {NAME_SUFFIX}:=
Text which appears on a name line after or behind the given and surname parts of a name.
e.g. Lt. Cmndr. Joseph /Allen/ Jr.
In this example Jr. is considered as the name suffix portion.

OBJE {OBJECT}:=
Pertaining to a grouping of attributes used in describing something. Usually referring to the data required to represent a multimedia object, such an audio recording, a photograph of a person, or an image of a document.

OCCU {OCCUPATION}:=
The type of work or profession of an individual.

PAGE {PAGE}:=
A number or description to identify where information can be found in a referenced work.

PEDI {PEDIGREE}:=
Information pertaining to an individual to parent lineage chart.

PHON {PHONE}:=
A unique number assigned to access a specific telephone.

PLAC {PLACE}:=
A jurisdictional name to identify the place or location of an event.

POST {POSTAL_CODE}:=
A code used by a postal service to identify an area to facilitate mail handling.

PROB {PROBATE}:=
An event of judicial determination of the validity of a will. May indicate several related court activities over several dates.

PROP {PROPERTY}:=
Pertaining to possessions such as real estate or other property of interest.

PUBL {PUBLICATION}:=
Refers to when and/or where a work was published or created.

QUAY {QUALITY_OF_DATA}:=
An assessment of the certainty of the evidence to support the conclusion drawn from evidence.

REFN {REFERENCE}:=
A description or number used to identify an item for filing, storage, or other reference purposes.

RELA {RELATIONSHIP}:=
A relationship value between the indicated contexts.

RELI {RELIGION}:=
A religious denomination to which a person is affiliated or for which a record applies.

REPO {REPOSITORY}:=
An institution or person that has the specified item as part of their collection(s).

RESI {RESIDENCE}:=
An address or place of residence that a family or individual resided.

RETI {RETIREMENT}:=
The event of ending one's occupational career.

RIN {REC_ID_NUMBER}:=
A number assigned to a record by an originating automated system that can be used by a receiving system to report results pertaining to that record.

ROLE {ROLE}:=
A name given to a role played by an individual in connection with an event.

ROMN {ROMANISED}:=
A romanised transcription of a superior text string.

SEX {SEX}:=
Indicates the sex of an individual; male, female, intersex or unknown.

SOUR {SOURCE}:=
The initial or original material from which information was obtained or (`HEAD.SOUR`) the system that created the GEDCOM file.

SPFX {SURN_PREFIX}:=
A name piece used as a non-indexing pre-part of a surname.

STAE {STATE}:=
A geographical division of a larger jurisdictional area (country), such as a province or state.

SUBM {SUBMITTER}:=
An individual or organization who contributes genealogical data to a file or transfers it to someone else.

SURN {SURNAME}:=
A family name passed on or used by members of a family.

TEXT {TEXT}:=
The exact wording found in an original source document.

TIME {TIME}:=
A time value in a 24-hour clock format, including hours, minutes, and optional seconds, separated by a colon (:). Fractions of seconds are shown in decimal notation.

TITL {TITLE}:=
A description of a specific writing or other work, such as the title of a book when used in a source context, or a formal designation used by an individual in connection with positions of royalty or other social status, such as Grand Duke.

TYPE {TYPE}:=
A further qualification to the meaning of the superior record. The value does not have any computer processing reliability. It is more in the form of a short one or two word note that should be displayed any time the associated data is displayed.

WIFE {WIFE}:=
A partner in a FAM (family group), often female, often partner to a man, and a biological, official or legal parent to each of the children of this couple.
The name of this record strongly suggests that the line value must identify a wife, but that is not the case; the relationship need not be a marriage, and the individual need not be female, it may identify a man in a gay relationship.

References

Same-Sex Marriage in GEDCOM

WILL {WILL}:=
A legal document treated as an event, by which a person disposes of his or her estate, to take effect after death. The event date is the date the will was signed while the person was alive. (See also PROB (probate), page 128.)

WWW {WEB}:=
World Wide Web address.

Appendix B ANSEL to Unicode Conversion

ANSEL to Unicode Conversion

There is more to conversion from ANSEL (or LANSEL) to Unicode than just the ANSEL to Unicode conversion tables. A major issue is that, in ANSEL, combining characters (modifiers) come before the base characters, while in Unicode they come after the base character.

After conversion from ANSEL to Unicode, the result should be normalised as appropriate for the platform. On Mac OS X, the string must be converted to Unicode Normal Form D (decomposed characters), while on Windows, it must be converted to Unicode Normal Form C (precomposed characters).
This normalisation step must not be skipped for either platform or normal form, as ANSEL text may contain both composed and decomposed characters.

ANSEL to Unicode Conversion Algorithm

1. Convert each ANSEL code point into the corresponding Unicode code point
2. mirror the positions of each base character and its modifiers (modifiers after instead of before base characters)
3. Convert to Unicode Normal Form C or Unicode Normal Form D, as appropriate for the platform

Unicode to ANSEL Conversion

Conversion from Unicode to ANSEL may seem simple; just perform the ANSEL to Unicode conversion step in reverse order:

1. Make sure the Unicode string is in Unicode Normal Form D
2. mirror the positions of each base characters and its modifiers (modifiers before instead of after base characters)
3. Replace Unicode code points with their ANSEL equivalents

However, Unicode to ANSEL conversion is more complicated than that. There are two issues that require attention:

- unsupported characters
- ANSEL's pre-composed characters

Unsupported Characters

First of all, ANSEL does not support all the same characters that Unicode supports, and does not feature an Replacement Character like Unicode does. The widespread convention for such cases is to replace the unsupported character by a question mark. When doing so, care must be taken to replace an unsupported character and all its modifiers by just one single question mark.

ANSEL does not support all the same combining characters as Unicode either. Generally speaking, when a modifier isn't supported, the Unicode character isn't supported, and should be replaced by a question mark.
It is arguably better, albeit inconsistent, to keep the base character if supported, and simply lose any unsupported modifiers.

ANSEL's Pre-Composed Characters

Special attention must be paid to ANSEL's pre-composed characters. For example, ANSEL does not contain an equivalent for *U+031B Combining Horn*, but it does contain equivalents for the pre-composed characters *U+101A Latin Capital Letter O with Horn* and *U+10AF Latin Capital Letter U with Horn*.

If your Unicode to ANSEL conversion fails to take the existence of ANSEL's pre-composed characters into account, your conversion will reduce those characters to their base character.

Unicode to ANSEL Conversion Algorithm

1. Make sure the Unicode string is in Unicode Normal Form D
2. De-normalise for pre-composed characters supported by ANSEL
3. mirror the positions of each base characters and its modifiers (modifiers before instead of after base characters)
4. Replace Unicode code points with their ANSEL equivalents
   - Replace unsupported characters with a question mark
   - Take care to replace an unsupported characters and all its modifiers by a single question mark
   - If a base character is supported, but a modifier is not, keep the base character, but leave off the modifier

The de-normalisation step is essential for taking advantage of ANSEL's pre-composed characters, and not ending up with just their base character instead.

Operating System Support

It is generally unwise to try and roll your own character set conversion functions, and best to rely on built-in functions for any character set handling and conversions. All major operating systems, operating environments, and several third-party libraries provide character set conversion functions, but no major systems and few libraries support ANSEL. However, all major operating systems provide functions for conversion of strings to Unicode Normal Forms:

- Microsoft Windows provides the `NormalizeString()` and `IsNormalizedString()` functions.
- Apple Mac OS X provides the `NSString()` function.
- Oracle Java provides the `Normalizer2` class, with multiple methods.
- Microsoft .NET provides the `String.Normalize()` method.
- Google Android provides the same `Normalizer2` class.
- Apple iOS provides the same `NSString()` function as Apple Mac OS X.

Best Practice for Genealogy Applications

- support import of ANSEL GEDCOM files
- do not export to ANSEL GEDCOM
  - always export to UTF-8 or UTF-16
- use built-in functions for Unicode normalisation

References

- ANSEL Administratively Withdrawn
- GEDCOM 5.5.1 Specification ANSEL Table
- LDS ANSEL versus LDS ANSEL
- ANSEL Alif & Ayn
- ANSEL / Unicode Conversion Tables
- ANSEL / Unicode Conversion Algorithms

Appendix C ANSEL Character Set

ANSEL Tables

ANSEL

The ANSEL character set (ANSI/NISO Z39.47-1985) was originally created in 1985 CE by the Standards Committee on Coded Character Sets for Bibliographic Information Interchange of America. ANSEL is an 8-bit character set that supports dozens of Western languages without requiring multiple, conflicting code pages to do so. ANSEL became widely used in library systems, but has effectively been replaced by Unicode. The ANSEL specification was last updated in 1993 CE, reaffirmed in 2003 CE, and administratively withdrawn in 2013 CE. ANSEL is mostly of historic interest now.

Left Hook

The ANSEL character 0xF7 left hook appears as “left hoof” in the ANSEL standard. That is an acknowledged typographical error. The actual name is “left hook”.

Replaced by Unicode

FamilySearch GEDCOM 4.0 (1989 CE) through 5.5.1 (1999 CE) recommend the use of ANSEL. The *GEDCOM 5.5.1 Annotated Edition* (2018 CE) already notes that the GEDCOM 5.5.1 rules are obsolete, that all genealogy applications should be Unicode-based and always export to Unicode.
GEDCOM 5.5.5 recommends that GEDCOM writers use Unicode instead of ANSEL.

Being phased out

Use of ANSEL in GEDCOM files is being phased out.
Use of the ANSEL character set is officially deprecated now.

New versions of existing systems *need not* and *should* provide the ability to export ANSEL-encoded GEDCOM file, but should be able to import ANSEL GEDCOM files, particularly ANSEL GEDCOM files created by a previous version of the same system.
New systems using GEDCOM 5.5.5 *must not* export ANSEL-encoded GEDCOM files, and *need not* support import of any ANSEL GEDCOM files.

LANSEL

Not ANSEL, but LANSEL

Technically, the tables provided in the GEDCOM specification aren't ANSEL tables, but LANSEL tables.
The FamilySearch tables leave ANSEL characters out and add extensions. Because these extensions are defined by the LDS (FamilySearch's parent company), they are known as LDS Extensions, and the resulting character set is known as LDS ANSEL, LANSEL for short.
GEDCOM keeps using the value `ANSEL` to specify the LANSEL character set for backward compatibility's sake.

Bad Tables

The LANSEL tables in GEDCOM 5.5.1 and earlier often do not display as intended, because FamilySearch relied on WordPerfect fonts that most users do not have installed. The *GEDCOM 5.5.1 Annotated Edition* contains LANSEL tables that does not rely on any particular font.

Multiple Versions

Different versions of FamilySearch GEDCOM contain different “ANSEL” tables. Different FamilySearch GEDCOM versions omit different ANSEL characters and add different extensions. Thus, FamilySearch actually defined different versions of LANSEL in different GEDCOM versions.
That creates unnecessary uncertainty about whether specific characters are supported or not.
The *GEDCOM 5.5.1 Annotated Edition* provides consolidated LANSEL tables as bonus information.

Consolidated Tables

The consolidated tables are a combination of the original tables in the ANSEL standard and the LANSEL tables that FamilySearch published in different GEDCOM versions. The consolidated tables contain all the ANSEL characters that FamilySearch omitted, as well as all the LDS extensions they added in different GEDCOM versions.

Recommended Tables

GEDCOM 5.5.5 provides the same consolidated LANSEL table as *GEDCOM 5.5.1 Annotated Edition* as the replacement for the tables provided in earlier versions of GEDCOM. Developers are recommended to not rely on those earlier tables, but simply use the consolidated tables provided here with previous versions of GEDCOM as well.

WordPerfect Codes

The LANSEL tables in GEDCOM 5.5.1 and earlier contain a column specifying wpcodes; the WordPerfect code page and code point for a character. That irrelevant information is no longer provided.

Unicode

The new LANSEL tables provide the Unicode code points and names that correspond with the LANSEL codes and the Unicode names (technically, the Unicode names should be ALL-UPPERCASE, but that does not help readability). It is recommended that developers use the Unicode names exclusively.

There is more to conversion from ANSEL or LANSEL to Unicode than just the ANSEL to Unicode conversion tables. See Appendix B *ANSEL to Unicode Conversion*, page 130.

Combining Characters

The graphics column of these tables, show the *actual* combining characters, not some non-combining character that approximates it. For example, the first character shown is U+0300 Combining Grave Accent, not U+0060 Grave Accent.
To show the combining characters correctly and to clearly show the relative position of the combining character, the table uses ◌ (U+25CC Dotted Circle) as a placeholder character.

ANSEL Non-spacing graphic characters

Hex	Dec	Graphic	Name	example of use	code point	Name
E0	224	◌̉	low rising tone mark	củi	U+0309	Combining Hook Above
E1	225	◌̀	grave accent	règle	U+0300	Combining Grave Accent
E2	226	◌́	acute accent	está	U+0301	Combining Acute Accent
E3	227	◌̂	circumflex accent	même	U+0302	Combining Circumflex Accent
E4	228	◌̃	tilde	niño	U+0303	Combining Tilde
E5	229	◌̄	macron	gājējs	U+0304	Combining Macron
E6	230	◌̆	breve	altă	U+0306	Combining Breve
E7	231	◌̇	dot above	żaba	U+0307	Combining Dot Above
E8	232	◌̈	umlaut (diaeresis)	öppna	U+0308	Combining Diaeresis
E9	233	◌̌	hacek	vždy	U+030C	Combining Caron
EA	234	◌̊	circle above (angstrom)	hår	U+030A	Combining Ring Above
EB	235	◌͡	ligature, left half	akademii͡a	U+FE20	Combining Ligature Left Half
EC	236	◌͜	ligature, right half	akademii͜a	U+FE21	Combining Ligature Right Half
ED	237	◌̓	high comma, off center	rozdel̓ovac	U+0315	Combining Comma Above Right
EE	238	◌̋	double acute accent	időszaki	U+030B	Combining Double Acute Accent
EF	239	◌̑	candrabindu	Ali̐ev	U+0310	Combining Cadrabindu
F0	240	◌̧	cedilla	ça	U+0327	Combining Cedilla
F1	241	◌̨	right hook, ogonek	vietą	U+0328	Combining Ogonek
F2	242	◌̣	dot below	teḍa	U+0323	Combining Dot Below
F3	243	◌̤	double dot below	khuṭbah	U+0324	Combining Diaeresis Below
F4	244	◌̥	circle below	Samskṛta	U+0325	Combining Ring Below
F5	245	◌̳	double underscore	G̳hulam	U+0333	Combining Double Low Line
F6	246	◌̲	underscore	s̲amar	U+0332	Combining Low Line
F7	247	◌̦	left hook	dārziņa	U+0326	Combining Comma Below
F8	248	◌̩	right cedilla	khọng	U+031C	Combining Left Half Ring Below
F9	249	◌̮	half circle below (upadhmaniya)	ḫumantuš	U+032E	Combining Breve Below
FA	250	◌͢	double tilde, left half	n͢galan	U+FE22	Combining Double Tilde Left Half
FB	251	◌͞	double tilde, right half	n͞galan	U+FE23	Combining Double Tilde Right Half
FC	252	◌̸	diacritic slash through char		U+0338	Combining Long Solidus Overlay
FD	253	⍰	*unused*		U+FFFD	Replacement Character
FE	254	◌̔	high comma, centered	gȅotermika	U+0313	Combining Comma Above
FF	255	⍰	*illegal*		U+FFFD	Replacement Character

ANSEL Spacing graphic characters

Hex	Dec	Graphic	Name	example of use	code point	Name
A0	160	⍰	*unused*		U+FFFD	Replacement Character
A1	161	Ł	slash L — uppercase	Łódź	U+0141	Latin Capital Letter L with Stroke
A2	162	Ø	slash O — uppercase	Øst	U+00D8	Latin Capital Letter O with Stroke
A3	163	Đ	slash D — uppercase	Đuro	U+0110	Latin Capital Letter D with Stroke
A4	164	Þ	thorn — uppercase	Þann	U+00DE	Latin Capital Letter Thorn
A5	165	Æ	ligature AE — uppercase	Ægir	U+00C6	Latin Capital Letter AE
A6	166	Œ	ligature OE — uppercase	Œuvre	U+0152	Latin Capital Ligature OE
A7	167	○′	mjagkij znak	fakul′tet	U+02B9	Modifier Letter Prime
A8	168	·	middle dot	novel·la	U+00B7	Middle Dot
A9	169	♭	musical flat	B♭	U+266D	Musical Flat Sign
AA	170	®	registered trademark	ABC®	U+00AE	Registered Sign
AB	171	±	plus or minus	A±B	U+00B1	Plus-Minus Sign
AC	172	Ơ	hook O - uppercase	BƠ	U+01A0	Latin Capital Letter O with Horn
AD	173	Ư	hook U - uppercase	XƯA	U+01AF	Latin Capital Letter U with Horn
AE	174	○’	alif	Un’yusho	U+02BC	Modifier Letter Apostrophe
AF	175	⍰	*unused*		U+FFFD	Replacement Character
B0	176	○‘	ayn	fa‘il	U+02BB	Modifier Letter Turned Comma
B1	177	ł	slash l— lowercase	rozbił	U+0142	Latin Small Letter L with Stroke
B2	178	ø	slash o— lowercase	høj	U+00F8	Latin Small Letter O with Stroke
B3	179	đ	slash d— lowercase	đavola	U+0111	Latin Small Letter D with Stroke
B4	180	þ	thorn— lowercase	þann	U+00FE	Latin Small Letter Thorn
B5	181	æ	ligature ae— lowercase	skæg	U+00E6	Latin Small Letter AE
B6	182	œ	ligature oe— lowercase	œuvre	U+0153	Latin Small Ligature OE
B7	183	○″	hard sign (tvjordyj znak)	ob″i͡avlenie	U+02BA	Modified Letter Double Prime
B8	184	ı	dotless i— lowercase	masalı	U+0131	Latin Small Letter Dotless I
B9	185	£	British pound	£5.00	U+00A3	Pound Sign
BA	186	ð	eth	verður	U+00F0	Latin Small Letter Eth
BB	187	⍰	*unused*		U+FFFD	Replacement Character
BC	188	ơ	hook o - lowercase	Sơ	U+01A1	Latin Small O with Horn
BD	189	ư	hook u - uppercase	Tự Đức	U+01B0	Latin Small U with Horn
BE	190	□	empty box (LDS-extension)		U+25A1	Empty Box
BF	191	■	black box (LDS-extension)		U+25A0	Black Box
C0	192	°	degree sign	10°C.	U+00B0	Degree Sign
C1	193	ℓ	script l	25 ℓ.	U+2113	Script Small L
C2	194	℗	phono copyright mark	Decca℗	U+2117	Sound recording copyright
C3	195	©	copyright mark	©1993	U+00A9	Copyright Sign
C4	196	♯	music sharp sign	D♯	U+266F	Music Sharp Sign
C5	197	¿	inverted question mark	¿Qué?	U+00BF	Inverted Question Mark
C6	198	¡	inverted exclamation mark	¡Esta!	U+00A1	Inverted Exclamation Mark
C7	199	⍰	*unused*		U+FFFD	Replacement Character
C8	200	⍰	*unused*		U+FFFD	Replacement Character
C9	201	⍰	*unused*		U+FFFD	Replacement Character
CA	202	⍰	*unused*		U+FFFD	Replacement Character
CB	203	⍰	*unused*		U+FFFD	Replacement Character
CC	204	⍰	*unused*		U+FFFD	Replacement Character
CD	205	e	e in middle of line (LDS extension)		U+0065	Latin Small Letter E
CE	206	o	o in middle of line (LDS extension)		U+006F	Latin Small Letter O
CF	207	ß	Ess Zed	Preußen	U+00DF	Latin Small Letter Sharp S

End of Specification

The end of the GEDCOM 5.5.5 specification.

Bonus Chapters

The Bonus Chapters are not part of the GEDCOM Specification proper.

GEDCOM Validation

Developers must make sure that the GEDCOM files exported by their application are valid GEDCOM files. There are several GEDCOM validators and some other tools available to help with that. With the exception of GedChk, these tools are fairly mature.

GEDCOM validators are themselves applications that may contain errors.
When in doubt, try multiple validation tools, and re-read the specification.

GEDCOM Validators

GedChk

GedChk is a free GEDCOM validator by FamilySearch. GedChk is a command-line application for MS-DOS, for which only version 0.9 Beta is available. GedChk supports GEDCOM 5.5 and 5.5.1. GedChk is available for FTP from the LDS website. GedChk is not well-behaved enough to run in a Windows DOS Box, but it does run in DOSBox, which is freely available for many operating systems.

VGedX

VGedX is a free GEDCOM validator created by Tim Forsythe, the creator of Gigatrees.
VGedX is a command-line application for Windows, complemented by a windowed configuration interface. VGedX supports GEDCOM 5.5, GEDCOM 5.5.1 and GEDCOM 5.6, and the user interface is available in multiple languages. VGedX is available for Windows, and requires 64-bit Windows 7 or later.

Chronoplex GEDCOM Validator

The Chronoplex GEDCOM Validator is a free product of Chronoplex (Andrew Hoyle), creators of Chronoplex Family Tree. The Chronoplex GEDCOM Validator is a Microsoft .NET application for Windows. The Chronoplex GEDCOM Validator supports GEDCOM 5.5, GEDCOM 5.5.1 and GEDCOM 5.6, and the user interface is available in multiple languages. The Chronoplex GEDCOM Validator requires Windows and Microsoft .NET.

GED-inline

GED-inline is a free on-line GEDCOM validator by Nigel Munro Parker.
GED-inline supports GEDCOM 5.5 and GEDCOM 5.5.1.

Additional Validation Tools

Behold

Behold is a commercial genealogy viewer by Louis Kessler. Behold's forgiving GEDCOM reader provides so many error and warning messages, that Behold be thought of as a GEDCOM validator in disguise.
Behold supports every version of GEDCOM, including GEDCOM 1.0. Behold requires Windows NT 4.0 or later.

GedPad

GedPad is free product of Nigel Button Software, makers of The Complete Genealogy Reporter. GedPad isn't a validator, but a GEDCOM editor with some validation features. GedPad's user-interface is a bit unusual and it does not support ASCII or ANSEL files, but it can be a handy tool

to have around when some GEDCOM file gives you trouble.

Genealogica Grafica

Genealogica Grafica is a free product by Tom de Neef, and the successor to his previous product, KStableau. Genealogica Grafica is a tool for creation of web charts and reports. On loading a GEDCOM file, it provides a report that includes GEDCOM errors and genealogy consistency checks. Genealogica Grafica support GEDCOM 5.5 and GEDCOM 5.5.1. Genealogica Grafica requires Windows.

References

- GEDCOM Validation
- DOSBox
- ftp://ftp.ldschurch.org/genealogy/GEDCOM/Utility/GedChk/
- VGedX, the Gigatrees GEDCOM Validator
- Chronoplex GEDCOM Validator
- GED-inline
- Behold
- GedPad
- Genealogica Grafica

GEDCOM Version Detection

Examine First

A GEDCOM reader cannot simply start reading a GEDCOM file. The various GEDCOM versions, character encodings and line terminators possible create a catch-22. A GEDCOM reader must start by *examining* the GEDCOM header, to discover the GEDCOM version, character encoding and line terminator used, before it can start *reading* the GEDCOM file - including the GEDCOM header. The GEDCOM reader can and should detect errors in the GEDCOM header.

Systems supporting GEDCOM need two separate things to read GEDCOM files; a GEDCOM header recognition routine and at least one GEDCOM reader. A typical system will have more than one GEDCOM reader, in support of multiple GEDCOM versions. The different readers take advantage of many of the same subroutines, such as those for character set conversion, but do parse GEDCOM files differently.

Choosing a GEDCOM Reader

The system chooses which GEDCOM reader to use for a particular GEDCOM file after examining the GEDCOM header and interpreting the results. A system should only try to read a particular GEDCOM file if it has a GEDCOM reader that supports its GEDCOM version and encoding. If a system does not support it, the system should admit implementation limitation, with a message such as “GEDCOM 4.0 is not supported. Please use GEDCOM 5.5 or later.”, and abort processing.

When the system picks a GEDCOM reader, it has not checked that the GEDCOM header is valid. It has merely examined the header to pick the right GEDCOM reader. It is up to the chosen GEDCOM reader to read the entire GEDCOM header and verify that the GEDCOM header is valid.

GEDCOM 5.5 & 5.5.1 versus GEDCOM 5.5.5

Many existing systems have a single GEDCOM reader that supports both GEDCOM 5.5 and GEDCOM 5.5.1. Many of these readers accept errors known to occur in third-party GEDCOM files, such as illegal tags and invalid GEDCOM headers. After more than two decades of systems producing sloppy GEDCOM 5.5 and 5.5.1 files, many GEDCOM 5.5.x readers are quite burdened by the complexity of supporting many known third-party errors.
Those GEDCOM 5.5 and 5.5.1 errors do not occur in GEDCOM 5.5.5 files, so a GEDCOM 5.5.5 reader need not and should not be burdened with that complexity.
Besides, GEDCOM 5.5.5 readers are not allowed to accept invalid GEDCOM.

Genealogy software developers *must not* extend their existing GEDCOM 5.5.1 reader to accept GEDCOM 5.5.5 files. Developers *must* create a new, separate GEDCOM 5.5.5 reader. GEDCOM 5.5.5 is a leaner, meaner GEDCOM, less complex and simpler to parse. This new reader should have much simpler and cleaner code, and be noticeable faster than the old GEDCOM 5.5.1 reader. The GEDCOM 5.5.5 reader *must* make sure the GEDCOM header is valid, and reject the ostensible GEDCOM file if it is not.

Newer GEDCOM Versions

If the HEAD.GEDC.VERS value is a higher value that the system does not recognise yet, the system *must* admit this implementation limitation, and abort. The system *must not* try to import the GEDCOM file until explicit support for the new GEDCOM version has been added.

GEDCOM 5.5.5

A GEDCOM 5.5.5 writer *must* create a correct GEDCOM header, and a GEDCOM reader *must not* accept invalid GEDCOM headers. Detection of GEDCOM 5.5.5 files is quite straightforward,